PHYS 135B Discussion 5

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I. $-\nabla^2 \frac{1}{r} = \nabla \cdot \frac{\mathbf{e}_r}{r^2} = 4\pi\delta^{(3)}(\mathbf{r})$ (HW4 Q6)

(a) Please check the expressions of Del in cylindrical and spherical coordinates. If $\mathbf{V} = V_r\mathbf{e}_r + V_\theta\mathbf{e}_\theta + V_\phi\mathbf{e}_\phi$ is a **continuously differentiable vector field** [1] , then the divergence of $\mathbf{V}$ in spherical coordinates can be written as

$$\nabla \cdot \mathbf{V} = \frac{1}{r^2}\frac{\partial(r^2 V_r)}{\partial r} + \frac{1}{r\sin\theta}\frac{\partial(\sin\theta V_\theta)}{\partial\theta} + \frac{1}{r\sin\theta}\frac{\partial(V_\phi)}{\partial\phi}.$$

Therefore,

$$\nabla \cdot \frac{\mathbf{e}_r}{r^2} = \frac{1}{r^2}\frac{\partial(r^2 \cdot \frac{1}{r^2})}{\partial r} = 0, \quad \text{when } r \neq 0.$$

(b) When $r = 0$, we resort to the volume integral of $\nabla \cdot \frac{\mathbf{e}_r}{r^2}$ over a sphere of radius R with the center sitting at the origin,

$$\begin{aligned}\int dV \nabla \cdot \frac{\mathbf{e}_r}{r^2} &= \oint d\mathbf{S} \cdot \frac{\mathbf{e}_r}{r^2} = \oint dS \mathbf{e}_r \cdot \frac{\mathbf{e}_r}{r^2} \\ &= \int_0^{2\pi} d\phi \int_0^{\pi} d\theta R^2 \sin\theta \frac{1}{R^2} \\ &= 4\pi = 4\pi \int dV \delta^{(3)}(\mathbf{r}).\end{aligned}$$

By comparing the first and the last term in the equation above, we end up with an important identity in EM,

$$\boxed{\nabla \cdot \frac{\mathbf{e}_r}{r^2} = -\nabla^2 \frac{1}{r} = 4\pi\delta^{(3)}(\mathbf{r})}.$$

[1] Note: This condition means that the equation for the divergence of a vector field fails in the region where the vector field is not differentiable. E.g. $\frac{\mathbf{e}_r}{r^2}$ is not differentiable at the origin, thus the divergence of it is not well-defined at $\mathbf{r} = 0$.

II. Average charge density over a ball(HW4 Q10)

(a) Generally, suppose there are N charged particles moving in space, different densities of physical quantities can be calculated by multiplying the physical quantity of each particle by the corresponding delta function, followed by taking a sum over all the particles. E.g.,

$$\text{Charge density: } \rho_c(\mathbf{r}) = \sum_i q_i \delta^{(3)}(\mathbf{r} - \mathbf{r}_i)$$

$$\text{Mass density: } \rho_m(\mathbf{r}) = \sum_i m_i \delta^{(3)}(\mathbf{r} - \mathbf{r}_i)$$

$$\text{Current density: } \mathbf{J}(\mathbf{r}) = \sum_i q_i \mathbf{v}_i \delta^{(3)}(\mathbf{r} - \mathbf{r}_i)$$

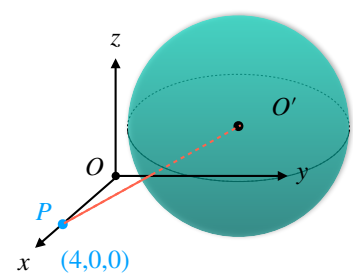
$$\text{Momentum density: } \mathbf{g}(\mathbf{r}) = \sum_i m_i \mathbf{v}_i \delta^{(3)}(\mathbf{r} - \mathbf{r}_i)$$

$$\ldots$$

Therefore, the charge density of an electron at $(4, 0, 0)$ is

$$\rho_c(\mathbf{r}) = -e\delta^{(3)}(\mathbf{r} - (4, 0, 0)).$$

(b) Now we want to calculate the average charge density within a sphere of radius $R = 2$ as a function of the position of the center of the sphere.



Notice that average charge density vanishes when $|O'P|$ is greater than the radius of the sphere since there is no electron inside. However, when $|O'P|$ is less than the radius of the sphere, the average density inside the sphere is $\bar{\rho} = -e/(\frac{4}{3}\pi R^3)$. Thus, using the Heaviside function $\theta(x)$

$$\bar{\rho}(\mathbf{r}) = \frac{-e}{\frac{32}{3}\pi}\theta(2 - |O'P|) = \frac{-e}{\frac{32}{3}\pi}\theta(2 - \sqrt{(x-4)^2 + y^2 + z^2})$$

III. $\nabla\frac{1}{|\mathbf{r}-\mathbf{r}'|}$ and $\nabla'\frac{1}{|\mathbf{r}-\mathbf{r}'|}$ (HW4 Q17)

In Cartesian coordinates, $\nabla = \mathbf{e}_x\partial_x+\mathbf{e}_y\partial_y+\mathbf{e}_z\partial_z$, and $\frac{1}{|\mathbf{r}-\mathbf{r}'|} = ((x-x')^2+(y-y')^2+(z-z')^2)^{-1/2}$. Therefore,

$$\begin{aligned}\nabla\frac{1}{|\mathbf{r}-\mathbf{r}'|} &= (\mathbf{e}_x\partial_x+\mathbf{e}_y\partial_y+\mathbf{e}_z\partial_z)\frac{1}{\sqrt{(x-x')^2+(y-y')^2+(z-z')^2}}\\ &= -\frac{(x-x')\mathbf{e}_x+(y-y')\mathbf{e}_y+(z-z')\mathbf{e}_z}{[(x-x')^2+(y-y')^2+(z-z')^2]^{3/2}}\end{aligned}$$

Similarly,

$$\begin{aligned}\nabla'\frac{1}{|\mathbf{r}-\mathbf{r}'|} &= (\mathbf{e}_x\partial'_x+\mathbf{e}_y\partial'_y+\mathbf{e}_z\partial'_z)\frac{1}{\sqrt{(x-x')^2+(y-y')^2+(z-z')^2}}\\ &= \frac{(x-x')\mathbf{e}_x+(y-y')\mathbf{e}_y+(z-z')\mathbf{e}_z}{[(x-x')^2+(y-y')^2+(z-z')^2]^{3/2}}\end{aligned}$$

We now obtain an important identity,

$$\boxed{\nabla\frac{1}{|\mathbf{r}-\mathbf{r}'|} = -\nabla'\frac{1}{|\mathbf{r}-\mathbf{r}'|}}.$$

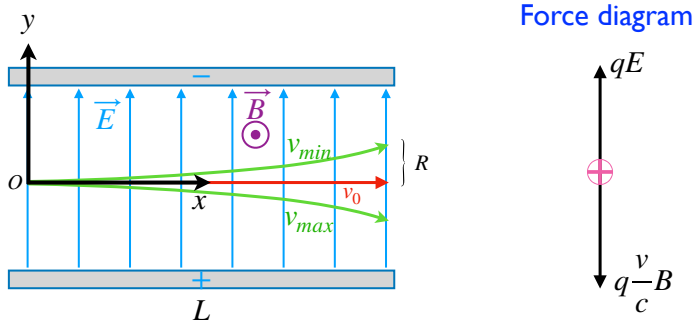
IV. Magnetic fields do no work on moving charges (HW4 Q18)

By definition, the work done by the magnetic field on a particle along path c is

$$\begin{aligned}W &= \int_c \mathbf{F}\cdot d\mathbf{r}\\ &= \int_c q(\mathbf{v}\times\mathbf{B})\cdot d\mathbf{r}\\ &= \int_{t_a}^{t_b} q(\mathbf{v}\times\mathbf{B})\cdot\mathbf{v}dt = \int_{t_a}^{t_b} q(\mathbf{v}\times\mathbf{v})\cdot\mathbf{B}dt = 0,\end{aligned}$$

where we have used the cyclic symmetry of scalar triple product. Intuitively, the magnetic force $\mathbf{F}$ is always perpendicular to $d\mathbf{r}$, thus $\mathbf{B}$ does no work on moving charges.

V. Wien filter(HW4 Q23)



Given that $R << L$, the direction of magnetic force doesn't change too much. We thus assume that it is always pointing downwards, leading to a net force along y-axis. As a result, particles experience projectile motion after they enter the filter. Then we may write down the kinematic equations for the upper green trajectory,

$$a = \frac{qE - qv_{min}B/c}{m} \qquad \text{(cgs)} \tag{1}$$

$$R = \frac{1}{2}at^2 \tag{2}$$

$$t = \frac{L}{v_{min}} \tag{3}$$

1. **An easy way**
   Notice that $\Delta v = v_{max} - v_{min}$ is much smaller than $v_0 = \frac{Ec}{B}$, the speed at which the particles travel through horizontally with zero net force. Therefore, we can approximate Eq**??** by

   $$t = \frac{L}{v_{min}} \approx \frac{L}{v_0}.$$

   Then, it can be shown that

   $$R = \frac{1}{2}\frac{qE - qv_{min}B/c}{m}\left(\frac{L}{v_0}\right)^2 \Rightarrow v_{min} = \frac{Ec}{B} - \frac{2mRE^2c^3}{qL^2B^3}.$$

   The kinematic equations for calculating v_{max} are almost the same except that the sign of R is flipped, i.e. $R \to -R$. As a result,

   $$v_{max} = \frac{Ec}{B} + \frac{2mRE^2c^3}{qL^2B^3},$$

   and the velocity range is

   $$\Delta v = v_{max} - v_{min} = \begin{cases} \frac{4mRE^2c^3}{qL^2B^3} & \text{(cgs)} \\ \frac{4mRE^2}{qL^2B^3} & \text{(SI)} \end{cases}$$

2. **A tedious way**

It can be shown from Eqs **??**, **??** and **??** that

$$R = \frac{1}{2}\frac{qE - qv_{min}B/c}{m}\left(\frac{L}{v_{min}}\right)^2$$

which can be further rearrange as a quadratic equation,

$$\lambda v_{min}^2 + v_{min} - \frac{Ec}{B} = 0$$

with $\lambda = \frac{2mRc}{qL^2B}$. The general solution to the equation above is

$$v_{min} = \frac{1}{2\lambda}\left(-1 + \sqrt{1 + \frac{4\lambda Ec}{B}}\right). \tag{4}$$

Newton's generalized binomial theorem:

$$\begin{aligned}(1+x)^r &= \sum_{k=0}^{\infty}\begin{pmatrix} r \\ k \end{pmatrix} x^k \\ &= 1 + rx + \frac{r(r-1)}{2!}x^2 + \frac{r(r-1)(r-2)}{3!}x^3 + ...\end{aligned}$$

where the binomial coefficients $\begin{pmatrix} r \\ k \end{pmatrix} = \frac{r(r-1)...(r-k+1)}{k!}$ and $|x| < 1$.

Given that $\frac{4\lambda Ec}{B}$ is a small dimensionless parameter, we now expand the square root in Eq**??** to the third order using the binomial theorem,

$$\begin{aligned}v_{min} &\approx \frac{1}{2\lambda}\left[-1 + 1 + \frac{1}{2}\cdot\frac{4\lambda Ec}{B} + \frac{\frac{1}{2}(\frac{1}{2}-1)}{2!}\left(\frac{4\lambda Ec}{B}\right)^2\right] \\ &= \frac{Ec}{B} - \frac{\lambda E^2c^2}{B^2} \\ &= \frac{Ec}{B} - \frac{2mRE^2c^3}{qL^2B^3}\end{aligned}$$

Similarly,

$$v_{max} = \frac{Ec}{B} + \frac{2mRE^2c^3}{qL^2B^3}.$$

We again obtain the same velocity range,

$$\Delta v = v_{max} - v_{min} = \begin{cases} \frac{4mRE^2c^3}{qL^2B^3} & \text{(cgs)} \\ \frac{4mRE^2}{qL^2B^3} & \text{(SI)} \end{cases}$$